

HFT Arbitrage in Commodities

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Commodities arbitrage has two regimes. Intra-exchange: fully hedged, favorable margin, and won on sub-microsecond latency. Inter-exchange: relative arbitrage, latency-sensitive, capital-intensive, held until mean reversion. How and why HFTs do both:

Intra-exchange

An underlying commodity on a futures exchange is more than a single contract. The full complex for a product includes:

- Outright futures for >20 different expirations
- Micro contracts with the same underlying but with smaller multipliers
- Calendar spreads between outrights
- Related intercommodity spreads

A futures exchange matching engine lists order books for all of these while also creating implied liquidity in both directions: implying spreads from outrights and implying outrights from spreads. Permutations of contracts and spreads present multiple arbitrage opportunities. Margin requirements for intra-exchange arbitrage trades are generally low given the offsets offered within the clearing house's risk system. These trades are highly competitive, and all the leading HFTs that participate have:

- Sub-microsecond tick-to-trade latency
- FPGAs for the entire trading system stack
- Logic which fires orders before all the bytes of ticks are processed
- Predictive models based on known market microstructure idiosyncrasies

Inter-exchange

Futures exchanges frequently list competing versions of contracts on the same underlying commodity. Unlike with equities where a company's stock is fully fungible regardless of which exchange it trades on (in all cases it is cleared at NSCC and held at DTC), a futures contract is specific to its exchange (DCM) and clearing house (DCO). An oil contract on two different exchanges will have incompatible clearing systems in addition to delivery differences.

The trading opportunities between exchanges are statistical or relative-value arbitrages rather than true arbitrages, but no less profitable. Traders need to pay for the best available market data path between venues, which may be microwave, millimeter-wave, or subsea fiber. Inter-

exchange trades tend to be less crowded than intra-exchange since considerable capital is required to maintain the spread between related products until it mean-reverts.

Whether through intra-exchange or inter-exchange trading, the profit earned by HFTs through arbitrage is compensation for keeping futures curves and related spreads consistent. The resulting curves allow commercial hedgers to accurately forecast storage, financing, and convenience costs across a variety of expirations, far into the future.

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